

GROUP A

CCTP CAPSTONE PRESENTATION

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PORSCHE TECHNOLOGY INNOVATION



Presenting to: Michael Steiner,
Member of Executive Board R&D of Porsche AG

Presented by: A Consulting

EXECUTIVE SUMMARY



Analysis

Despite new electric models, there will be lots of combustion engine vehicles on the road for decades to come.

eFuels are far too expensive for the public to afford.

Question

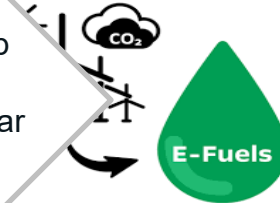
Should Porsche continue to invest in eFuels? Or transition fully into EVs?

Recommendations

Continue to develop electrical car models



Continue to develop eFuel and sell as admixture with regular fuel



Impacts

Transition more than 80% of new vehicle sales to full-electric models by 2030.

Make sustainability affordable by 2026 and achieve 100% blending rate by 2050.



Despite new electric models, there are still many vehicles with combustion engines on the road.

“Today, there are around 1.3 billion vehicles with combustion engines on the road worldwide. And, according to the forecasts, this number is not going to fall significantly in the next 15 years — despite the ramp-up of electromobility.”





Despite new electric models, there are still many vehicles with combustion engines on the road.

“Today, there are around 1.3 billion vehicles with combustion engines on the road worldwide. And, according to the forecasts, this number is not going

There are over a billion vehicles with combustion engines on the road despite the push for EVs. They will be on the road for years to come.

80%



By 2030

Remaining

20%

Engine Vehicles

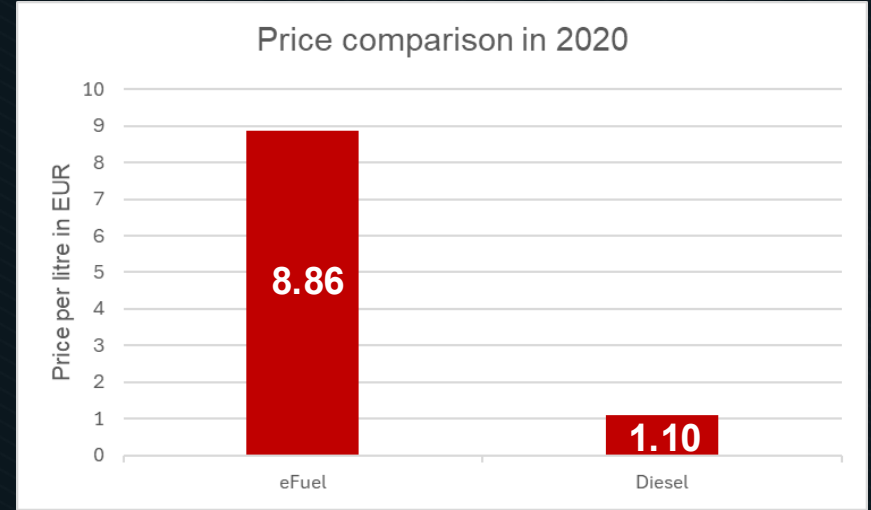
1.3 Bil

eFuel is far too expensive.



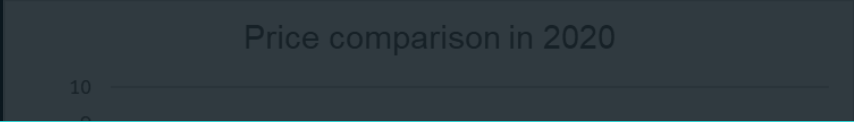
130,000 litres

By 2023



(Statista)

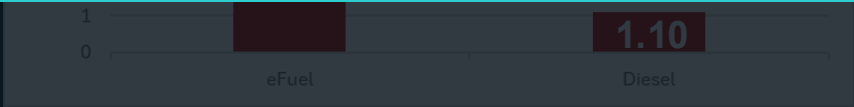
eFuel is far too expensive.



eFuel is sustainable, but at €8.86 per litre (as of November 2020), it's far too expensive compared to diesel.

130,000 litres

By 2023



(Statista)



Should Porsche continue to invest
in **eFuels**, or transition **fully** into
EVs?



Continue EV development to achieve 2030 goal

New Rivals



High-performance
battery technology

Competitive Edge

CELLFORCE

GROUP14

Advanced
vehicle dynamics



Results

More than 80% of all
new vehicles as
purely electric

Take a leap towards
sustainability and
gain a competitive
edge with the
development of
advanced luxury
EVs

Continue EV development to achieve 2030 goal

New Rivals



Competitive Edge



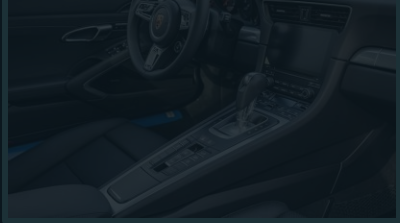
Results

More than 80% of all

Transition more than 80% of new vehicle sales to full-electric models by 2030.



Advanced
vehicle dynam



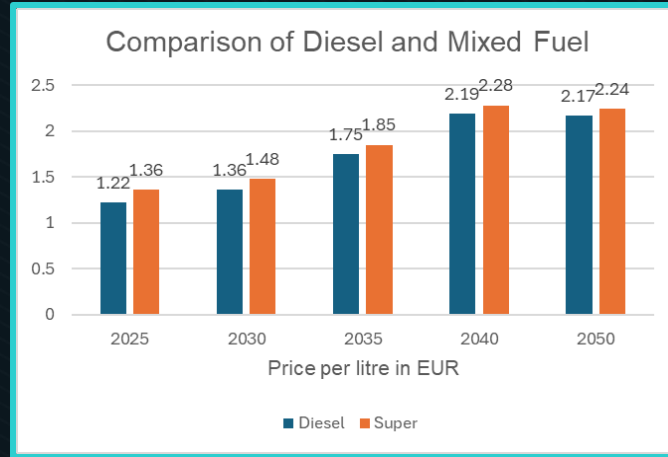
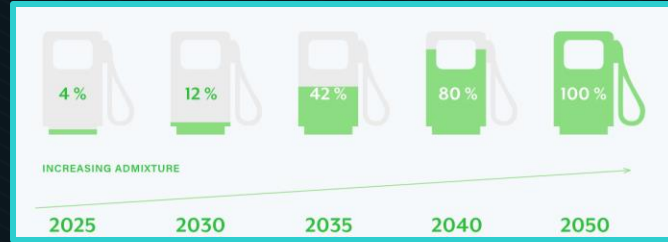
edge with the
development of
advanced luxury
EVs

Continue the development of eFuel and sell as admixture with regular fuel.



Problem

eFuel is too expensive for customers



Results

Increased quantities of eFuels being added gradually to conventional fuels

eFuel price remains at consistently affordable rate for consumers by 2050

Source: Prognos AG, Fraunhofer UMSICHT and DBFZ (2018): Status and perspectives of liquid energy sources in the Energy transition

Continue the development of eFuel and sell as admixture with regular fuel.

Ask



Results

Increased quantities
5.5 million tonnes

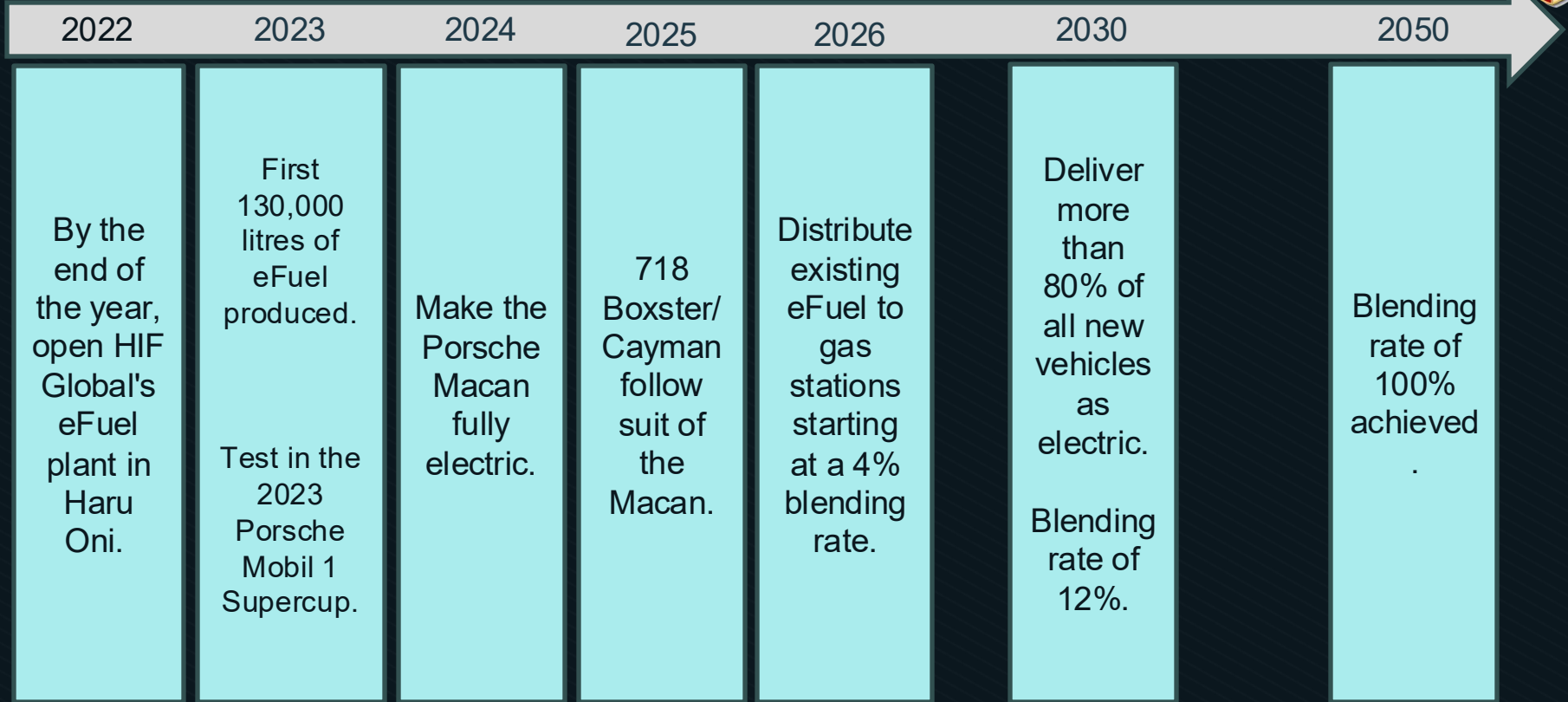
Through gradual increase in the blending rate starting from 2026, sustainability will remain an affordable option for all. By 2050, a blending rate of 100% will be reached.



affordable rate for
consumers by 2050

Source: Prognos AG, Fraunhofer UMSICHT and DBFZ (2018): Status and perspectives of liquid energy sources in the Energy transition

IMPLEMENTATION TIMELINE



Executive Summary

Analysis

Recommendation

Implementation

Conclusion



RISKS AND MITIGATION PLAN

Risk	Production will not go as planned and eFuel will remain expensive.	Lack of knowledge and interest in eFuels.	Concerns of loyal customers on electrical sports car
Impact			
Likelihood			
Mitigation	Adjust the blending rate as necessary to ensure the price remains affordable.	Educate and advertise eFuel at the Porsche Mobil 1 Supercup events.	Proving the efficiency of battery technology and performance quality standard and advertise the mixed eFuel

CONCLUSION



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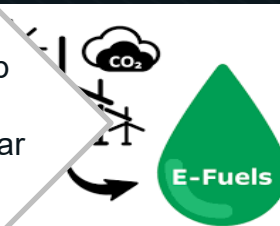
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EV COMPETITOR ANALYSIS

TESLA

- Electric-only car producer
- Strong brand image
- Supercharger network is convenient

XPENG MOTORS

- Offers 4 fully electric and highly software-oriented models
- Is rapidly expanding to international markets

BYD AUTO

- Diverse range of electric vehicles
- Invests in R&D for autonomous driving and battery technology

NIO INC

- Created 8 electric lines ranging from sedans to SUVs
- Sells premium electric vehicles

JAGUAR

- Known for modern luxury vehicles
- Wants to reach carbon net zero by 2039

BENEFITS OF EFUEL

Sustainable

- Uses renewable energy resources
- Can achieve carbon neutrality
- Can be the perfect substitute for a gas/diesel car

Fits Consumer Wants

- Longer Driving ranges
- Sounds and feels like a gas/diesel car
- Can be more eco-friendly due to their carbon-neutral characteristics

Compatible

- Can be used in existing internal combustion engines
- Does not require extensive modifications to infrastructure
- Efuels can be an immediate solution to regions where EV infrastructure is still developing.

PORSCHE EV TECHNOLOGY

Battery Technology

- Develops next-generation batteries that offer greater range and faster charging time.
- Partners with Cellforce Group and Group 14 to further develop high-performance batteries with high-charging capabilities.



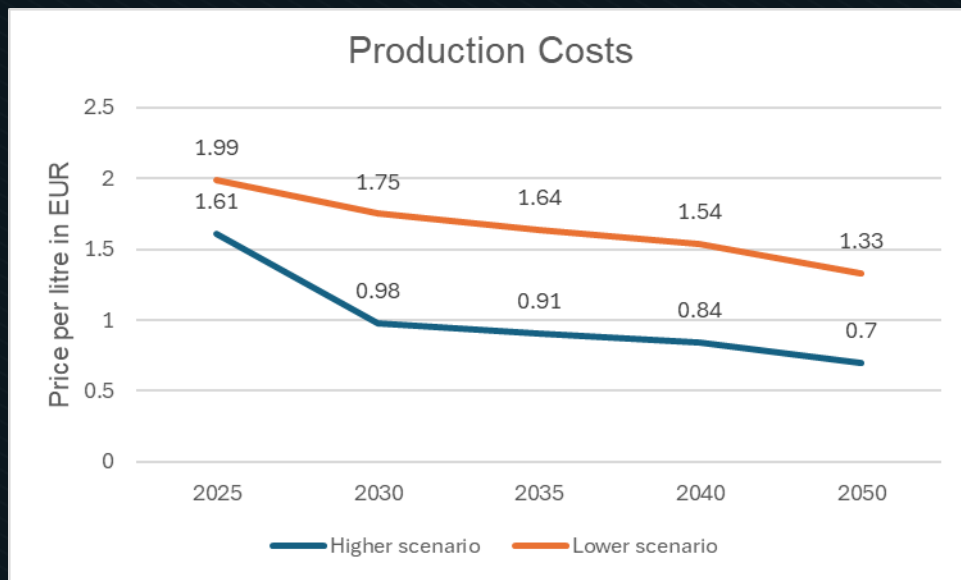
Why Partner with Cellforce?

- Battery is more compact, reduces internal resistance
- Absorbs more energy during energy recovery
- Allows improved performance and allows different driving modes and customizations (normal, sport, range)

APPENDIX



Estimated Production Costs of Admixture of eFuel



Source: Prognos AG, Fraunhofer UMSICHT and DBFZ (2018): Status and perspectives of liquid energy sources in the Energy transition